

A PROJECT REPORT ON
Multi-Priority Emergency Vehicle RF Beacon Signal Circuit with
Automatic Reset Logic

*Submitted in partial fulfillment of the requirements for the award of
degree of*

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project report entitled “ **Multi-Priority Emergency Vehicle RF Beacon Signal Circuit with Automatic Reset Logic** ” is a bonafide record submitted by AKULA ASHRITH, SALENDRA DAYA SAGAR, BODA ROHITH REDDY, VELPULA RAMA KRISHNA Department of Electronics and communication Engineering, Anurag University and is submitted in partial fulfillment for the award of Degree of Bachelor of Technology in “Electronics and Communication Engineering” for the year 2025- 2026. The work reported herein does not form part of any other thesis on which a degree has been awarded earlier.

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ACKNOWLEDGEMENT

It is indeed a great pleasure to thank all the people who had given support in completion of the project.

We would like to express our deep sense of gratitude and respect to our project guide **Dr. NANDOORI SRIKANTH** Associate Professor, Department of Electronics and Communication Engineering, for his encouragement, excellent guidance and kind support during our project, without which it would not have been possible to complete this work. We are thankful to him for being highly cooperative throughout our project work.

We extremely grateful to **Dr. Harikrishna Kamatham** , Professor & Head of the Department of Electronics and Communication Engineering, for her encouragement and support throughout the project.

Special thanks are dedicated to **Prof. V. Vijaya Kumar** , Dean School of Engineering, who helped us for the successful completion of our project.

We are extremely grateful to **Dr. Palla Rajeshwar Reddy**, Chairman, Anurag Group of Institutions, who encouraged and helped us for the successful completion of our project.

It is our pleasure to express thanks to all the teaching and non-teaching staff of Electronics and Communication Engineering Department for their excellent monitoring and their suggestions that helped in the successful completion of course work for B.Tech, Program.

Finally, we thank our parents and the Almighty, whose divine grace provided us the opportunity to do our project work according to our wish.

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DECLARATION

We hereby declare that the project report entitled, “**Image Multi-Priority Emergency Vehicle RF Beacon Signal Circuit with Automatic Reset Logic**”, is the work done by us and submitted for the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering, under the guidance of

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ABSTRACT

Urban traffic congestion has become one of the most critical challenges in modern smart cities. Emergency vehicles such as ambulances, fire engines, and police vehicles frequently experience delays at traffic intersections due to conventional fixed-time signal control systems. Even minimal delays can result in severe consequences, particularly in life-threatening medical emergencies.

This project proposes a **Multi-Priority Emergency Vehicle RF Beacon Signal Circuit with Automatic Reset Logic**, designed to provide dynamic traffic signal priority to emergency vehicles using secure Radio Frequency (RF) communication. The system operates using hardware-based encoding and decoding techniques without relying on microcontrollers or internet connectivity.

The emergency vehicle is equipped with a transmitter module that encodes priority data using a parallel-to-serial encoder IC. This encoded data is transmitted via a 433 MHz RF module using Amplitude Shift Keying (ASK) modulation. At the traffic intersection, a receiver module captures the signal and decodes it using a serial-to-parallel decoder IC.

A digital logic circuit resolves conflicts between multiple emergency vehicles using predefined priority levels:

- Ambulance – Highest Priority
- Fire Engine – Medium Priority
- Police – Lowest Priority

An automatic reset mechanism ensures that once the emergency vehicle clears the intersection, the signal returns to normal operation. This reset is implemented using latch-based memory circuits and timing control mechanisms.

The system integrates analog communication principles, digital logic design, RF transmission, and electronic interfacing to create a reliable and low-cost Smart City traffic priority solution. Experimental validation confirms correct priority execution, reliable RF communication, and successful automatic restoration of normal traffic flow.

CONTENTS

CERTIFICATE	ii
ACKNOWLEDGEMENT	iii
DECLARATION	iv
ABSTRACT	v
LIST OF FIGURES	x
LIST OF TABLES	xi
LIST OF ABBREVIATIONS	xii

CHAPTER-I

INTRODUCTION	1-5
1.1 Overview OF Emergency Vehicle Traffic Control	1
1.2 Need for Emergency Vehicle Priority Systems	2
1.3 Basic Concepts of RF Communication	3
1.4 Role of RF Technology in Traffic Management	3
1.5 Problem statement	4
1.6 Objectives of the project	4
1.7 Scope of the project	4
1.8 Organization of the Project	3

CHAPTER-II

LITERATURE REVIEW	6-7
2.1.Introduction	6
2.2.Review of existing Traffic Control techniques	6
2.2.1 Sensor-Based Traffic Control Systems	6
2.2.2 Sensor-Based Traffic Control Systems	6
2.2.3 Sensor-Based Traffic Control Systems	6

2.3.Review of RF-Based Traffic Systems	7
2.4.Summary of related research papers	7
2.5.Limitations of previous works	7
2.6.Research gap identified	7

CHAPTER-III

EXISTING METHODS **6-7**

3.1.Introduction	8
3.2. Conventional Traffic Signal Systems	8
3.3. Sensor-Based Traffic Systems	9
3.4. Camera-Based Traffic Monitoring Systems	10
3.5. Infrared-Based Emergency Vehicle Systems	10
3.6. Limitations of Existing Methods	11

CHAPTER-IV

PROPOSED METHODOLOGY **12-15**

4.1 Introduction	12
4.2 System architecture	12
4.3 RF Transmitter Unit	13
4.4 RF Receiver Unit	14
4.5 Priority logic Circuit	14
4.6 Automatic Reset Circuit	15

CHAPTER-V

HARDWARE DESIGN	16-19
5.1 Introduction	16
5.2 RF Transmitter Module	16
5.3 RF Receiver Module	17
5.4 Push Button Switch	17
5.5 Logic Gates	17
5.6 555 Timer IC	18
5.7 Traffic light Module	19
5.8 Power Supply Unit	19

CHAPTER-VI

SIMULATION RESULTS	20
6.1 Introduction	20
6.2 Testing Procedure	20
6.3 Experimental Observations	20
6.4 Performance Analysis	20

CHAPTER-VII	
ADVANTAGES AND APPLICATIONS	21
7.1 Advantages of the Proposed Method	21
7.2 Applications	21
CHAPTER-VIII	
CONCLUSION AND FUTURE SCOPE	22
8.1 Summary Of Work	22
8.2 Conclusion drawn from results	22
8.3 Limitations of the current system	22
8.4 Future scope	22
REFERENCES	23-24
APPENDIX	25-27

LIST OF FIGURES

Figure No.	Name of the Figure	Page No.
Fig. 4.2	System Architecture	13
Fig. 4.4	RF Transmitter & Receiver	14
Fig. 4.5	Priority Logic Circuit	15
Fig. 5.2	RF Transmitter Module	16
Fig. 5.3	RF Receiver Module	17
Fig. 5.4	Push Button switch	17
Fig.5.5	Logic Gates	18
Fig.5.6	555 Timer IC	18
Fig.5.7	Traffic Light Module	19
Fig.5.8	Power Supply Unit	19
Fig. B	Circuit Diagram	26
Fig. D	Traffic Signal Timing Waveform	27
Fig. E	Hardware Setup	

LIST OF TABLES

Table No.	Name of the Table	Page No.
Table 1.1	Problems in Conventional Traffic Systems	2
Table 1.8	Organization of the Project	5
Table 2.4	Summary of Research Papers	7
Table 3.2	Conventional Traffic signal systems	9
Table 4.5	Priority Logic Circuit	14
Table 6.3	Experimental observations	20
Table A	List of components	25
Table C	Truth Table for Priority Logic	26

LIST OF ABBREVIATIONS

S. No	Abbreviation	Full Form	Description
1	RF	Radio Frequency	Wireless communication technology used to transmit signals between emergency vehicle and traffic signal.
2	TX	Transmitter	Device that sends RF signals from the emergency vehicle.
3	RX	Receiver	Device that receives RF signals at the traffic signal controller.
4	IC	Integrated Circuit	Electronic circuit fabricated on a semiconductor chip used for logic and control operations.
5	PCB	Printed Circuit Board	Board used to mount and connect electronic components in the circuit.
6	LED	Light Emitting Diode	Indicator light used to represent traffic signals such as red, yellow, and green.
7	VCC	Voltage at the Common Collector	Positive supply voltage provided to electronic components.
8	GND	Ground	Reference point in the electrical circuit for current return path.
9	AND	Logical AND Gate	Digital logic gate that produces high output only when all inputs are high.
10	OR	Logical OR Gate	Digital logic gate that produces high output when at least one input is high.
11	NOT	Logical NOT Gate	Digital logic gate that inverts the input signal.
12	CMOS	Complementary Metal Oxide Semiconductor	Technology used for constructing integrated circuits.
13	TTL	Transistor-Transistor	Logic family used in digital

S. No	Abbreviation	Full Form	Description
		Logic	electronics circuits.
14	Hz	Hertz	Unit of frequency representing cycles per second.
15	kHz	Kilohertz	Frequency unit equal to 1000 Hertz.
16	MHz	Megahertz	Frequency unit equal to one million Hertz.
17	V	Volt	Unit of electrical potential difference.
18	A	Ampere	Unit of electric current.
19	Ω	Ohm	Unit of electrical resistance.
20	mA	Milliamper	Unit of current equal to one thousandth of an ampere.
21	RF Module	Radio Frequency Module	Module used for wireless communication between transmitter and receiver.
22	Logic Circuit	Digital Logic Circuit	Circuit that processes input signals using logic gates to control outputs.
23	Priority Logic	Priority Control Circuit	Circuit used to give higher priority to emergency vehicles.
24	Reset Circuit	Automatic Reset Circuit	Circuit used to return the traffic signal system to normal operation after emergency vehicle passes.
25	Traffic Control Unit	Traffic Signal Controller	Electronic system that controls the operation of traffic lights.

CHAPTER-I

INTRODUCTION

1.1 Overview of Emergency Vehicle Traffic Control

Urbanization Modern urban traffic systems face significant challenges due to increasing vehicle density and congestion. One of the most critical issues occurs when emergency vehicles such as ambulances, fire trucks, and police vehicles are delayed in traffic. These vehicles require immediate right of way to reach accident locations, hospitals, or emergency situations as quickly as possible. Even a small delay in emergency response can result in severe consequences, including loss of human life.

Traditional traffic signal systems operate on fixed timing mechanisms that do not adapt to emergency situations. In such systems, traffic lights continue to operate according to pre-programmed cycles regardless of the presence of emergency vehicles. As a result, emergency vehicles often become stuck at intersections waiting for signals to change, which wastes valuable time.

To address this issue, intelligent traffic control mechanisms are required. One effective solution is the use of a wireless emergency beacon signal system that allows emergency vehicles to communicate with traffic signals. When an emergency vehicle approaches an intersection, it can transmit a signal to the traffic signal controller to automatically give it priority.

This project proposes a **Multi-Priority Emergency Vehicle RF Beacon Signal Circuit with Automatic Reset Logic**, which enables emergency vehicles to control traffic lights using radio frequency communication. The system detects signals transmitted from emergency vehicles and automatically changes the traffic signal to green in the corresponding direction, allowing the emergency vehicle to pass safely and quickly.

1.2 Need for Emergency Vehicle Priority Systems

Emergency vehicles require immediate passage through traffic intersections. However, traffic congestion often prevents them from moving efficiently. Drivers may attempt to give way to emergency vehicles, but this is not always possible in crowded roads.

Some common challenges faced by emergency vehicles include:

- Heavy traffic congestion
- Fixed traffic signal timings
- Lack of automated emergency vehicle detection
- Dependence on manual traffic control

These problems increase emergency response time and reduce the efficiency of emergency services.

An automated traffic control system that prioritizes emergency vehicles can significantly reduce these delays. By allowing emergency vehicles to control traffic signals, such systems ensure faster response times and improved emergency management.

Problem	Description
Fixed signal timing	Traffic signals operate on fixed cycles
Traffic congestion	Vehicles accumulate at intersections
Manual control	Traffic police intervention required
No priority system	Emergency vehicles wait like regular vehicles

Table1.1 Problems in Conventional Traffic Systems

1.3 Basic Concepts of RF Communication

Radio Frequency (RF) communication is a wireless communication technique that uses electromagnetic waves to transmit signals through the air. RF communication is widely used in many applications including remote controls, wireless networks, and communication systems.

An RF communication system typically consists of:

- RF transmitter
- RF receiver
- Antenna
- Encoder and decoder circuits

The RF transmitter converts electrical signals into radio waves and transmits them through an antenna. The RF receiver captures the radio waves and converts them back into electrical signals.

RF communication offers several advantages:

- Long communication range
- Reliable wireless transmission
- Low power consumption
- Easy circuit implementation

These features make RF communication suitable for emergency vehicle traffic management systems.

1.4 Role of RF Technology in Traffic Management

RF technology enables wireless communication between emergency vehicles and traffic signal controllers. When an emergency vehicle approaches an intersection, it sends an RF signal to the traffic signal system.

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The traffic signal controller detects the signal and changes the signal phase accordingly. This allows the emergency vehicle to pass through the intersection without delay.

Advantages of RF-based traffic control systems include:

- Wireless operation
- Reduced infrastructure requirements
- Low implementation cost
- Improved system reliability

1.5 Problem Statement

Emergency vehicles frequently experience delays at traffic signals due to the absence of automated traffic priority systems. Conventional traffic signals operate on fixed timing cycles and cannot detect emergency vehicles.

Therefore, there is a need to design an intelligent traffic control system that allows emergency vehicles to transmit signals to traffic signal controllers and obtain priority passage.

1.6 Objectives of the project

The main objectives of the project are:

- To design a wireless RF-based emergency vehicle signal system
- To enable communication between emergency vehicles and traffic signals
- To implement multi-priority emergency vehicle detection
- To automatically control traffic lights based on emergency vehicle signals
- To implement an automatic reset mechanism after the emergency vehicle passes

1.7 Scope of the Project

The proposed system focuses on designing a prototype hardware circuit that demonstrates emergency vehicle priority traffic control using RF communication.

The system can be expanded in the future for smart city traffic management systems.

1.8 Organization of the Project

The project report is organized into eight chapters:

Chapter	Description
Chapter 1	Introduction
Chapter 2	Literature Review
Chapter 3	Existing Methods
Chapter 4	Proposed Methodology
Chapter 5	Hardware Design
Chapter 6	Experimental Results
Chapter 7	Advantages and Applications
Chapter 8	Conclusion and Future Scope

Table 1.8: Organization of the Project

CHAPTER-II

LITERATURE REVIEW

2.1 Introduction

A literature survey involves studying previously developed systems and research related to traffic signal control and emergency vehicle prioritization. Various researchers have proposed intelligent traffic systems using wireless communication, sensors, and microcontrollers.

This chapter reviews some of the important research contributions in this field.

2.2 Review of Existing Traffic Control Techniques

2.2.1 Sensor-Based Traffic Control Systems

Sensor-based traffic control systems use sensors such as inductive loop sensors and infrared sensors to detect vehicles on the road. These sensors measure traffic density and adjust signal timings accordingly.

However, these systems cannot distinguish emergency vehicles from regular vehicles.

2.2.2 Sensor-Based Traffic Control Systems

Camera-based systems use image processing techniques to monitor traffic conditions. Cameras installed at intersections capture images of vehicles and analyze traffic density.

Although effective, these systems are expensive and require complex processing algorithms.

2.2.3 Sensor-Based Traffic Control Systems

Infrared communication systems use infrared transmitters installed in emergency vehicles. When the vehicle approaches an intersection, it sends an infrared signal to the traffic signal controller.

However, infrared signals require direct line-of-sight and can be blocked by obstacles.

2.3 Review of RF-Based Traffic Systems

RF communication has been widely used in traffic management systems due to its reliability and long communication range. RF-based systems allow emergency vehicles to send signals to nearby intersections.

These systems provide faster signal detection and improved system reliability.

2.4 Summary of Research Papers

Author	Year	Method	Limitation
Smith et al.	2019	RF traffic control	Limited coverage
Kumar et al.	2020	Sensor-based system	Cannot detect emergency vehicles
Zhang et al.	2022	IoT traffic system	High cost

Table 2.4: Summary of Research Papers

2.5 Limitations of Previous Works

- High implementation cost
- Complex system design
- Limited communication range

2.6 Limitations of Previous Works

Most existing traffic control systems do not provide a simple, low-cost RF-based system with multi-priority emergency vehicle detection and automatic reset functionality.

CHAPTER-III

EXISTING METHODS

3.1 Introduction

Traffic congestion has become a major concern in modern urban areas. As the number of vehicles increases, traffic intersections experience heavy congestion, which leads to delays and inefficient traffic movement. Conventional traffic control systems are designed to regulate vehicle flow using predefined signal timing sequences. However, these systems do not consider special situations such as emergency vehicle movement. Emergency vehicles such as ambulances, fire trucks, and police vehicles require immediate access through traffic intersections to reach their destination quickly. Unfortunately, traditional traffic signal systems treat all vehicles equally and do not provide priority to emergency vehicles.

Over the years, several methods have been developed to improve traffic control systems. These methods include conventional traffic signal systems, sensor-based traffic control systems, camera-based traffic monitoring systems, and infrared-based emergency vehicle detection systems.

Each of these systems has certain advantages and limitations. Understanding these existing methods is important in order to identify their shortcomings and develop improved traffic control solutions.

3.2 Conventional Traffic Signal Systems

Traditional traffic signals operate on fixed timing cycles. In these systems, traffic lights change from red to green and yellow according to predetermined time intervals. These systems are commonly used in urban intersections because they are simple and inexpensive to implement.

The typical sequence of a conventional traffic signal system is:

1. Red Signal – Vehicles must stop.
2. Green Signal – Vehicles are allowed to move.
3. Yellow Signal – Vehicles prepare to stop.

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Signal	Duration
Red	30 seconds
Yellow	5 seconds
Green	25 seconds

Table 3.2:Conventional Traffic signal systems

Although these systems work effectively under normal traffic conditions, they do not adapt to changes in traffic density or the presence of emergency vehicles. As a result, emergency vehicles may be forced to wait at red signals.

Limitations of Conventional Traffic Signals

- Fixed signal timing regardless of traffic conditions
- No mechanism for detecting emergency vehicles
- Inefficient traffic flow during peak hours
- Increased waiting time for vehicles

3.3 Sensor-Based Traffic Systems

Sensor-based systems detect vehicle density using sensors installed on roads to detect vehicle density. Commonly used sensors include inductive loop sensors, infrared sensors, and ultrasonic sensors.

These sensors detect the presence of vehicles and send signals to the traffic controller. The traffic signal timing is then adjusted based on the number of vehicles waiting at the intersection.

Advantages of Sensor-Based Systems

- Dynamic adjustment of signal timings
- Reduced traffic congestion
- Improved traffic flow

Limitations

- Cannot differentiate emergency vehicles from regular vehicles
- High installation cost
- Sensors may malfunction due to environmental conditions

3.4 Camera-Based Traffic Monitoring Systems

Camera-based traffic control systems use surveillance cameras and image processing techniques to monitor traffic conditions. The captured images are analyzed using computer algorithms to determine traffic density.

These systems are widely used in smart city traffic management systems.

Advantages

- Real-time traffic monitoring
- Accurate vehicle detection
- Traffic violation detection

Limitations

- High installation cost
- Requires advanced processing systems
- Sensitive to weather conditions such as fog or rain

3.5 Infrared-Based Emergency Vehicle Systems

Infrared-based systems use infrared transmitters installed in emergency vehicles. When the vehicle approaches an intersection, it sends an infrared signal to the traffic signal controller.

The controller detects the signal and changes the traffic signal accordingly.

Advantages

- Faster emergency vehicle detection
- Automatic traffic signal control

Limitations

- Infrared signals require line-of-sight communication
- Signals can be blocked by obstacles
- Limited communication range

3.6 Limitations of Existing Methods

Although several traffic control systems have been developed, most of them suffer from the following limitations:

- Lack of reliable emergency vehicle detection

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- High system cost
- Complex system design
- Limited communication range

These limitations highlight the need for a simple and reliable traffic control system that can prioritize emergency vehicles efficiently.

CHAPTER-IV

PROPOSED METHODOLOGY

4.1 Introduction

The proposed system aims to provide priority access to emergency vehicles at traffic intersections using RF communication technology. The system consists of two main units: an RF transmitter unit installed in emergency vehicles and an RF receiver unit installed at traffic signal controllers.

When an emergency vehicle approaches an intersection, it transmits a wireless signal using the RF transmitter. The RF receiver detects this signal and activates the traffic signal control circuit. The system then changes the traffic light to green for the direction of the emergency vehicle.

The system also includes a priority logic circuit that determines which emergency vehicle receives priority when multiple signals are received simultaneously. Additionally, an automatic reset circuit ensures that traffic signals return to normal operation after the emergency vehicle passes.

4.2 System Architecture

The system architecture consists of the following main blocks:

- RF transmitter unit
- RF receiver unit
- Priority logic circuit
- Traffic light control circuit
- Automatic reset circuit

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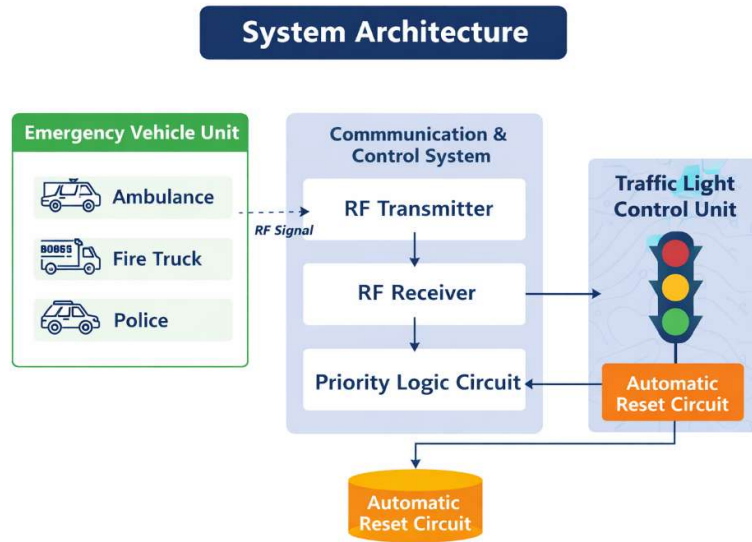


Fig 4.2: System Architecture

Working Principle

1. Emergency vehicle activates RF transmitter.
2. RF signal is transmitted wirelessly.
3. RF receiver detects the signal.
4. Priority logic circuit processes the signal.
5. Traffic signal changes to green.
6. Emergency vehicle passes intersection.
7. Automatic reset circuit restores normal traffic operation.

4.3 RF Transmitter Unit

The RF transmitter unit is installed inside emergency vehicles. It consists of a push button switch connected to an RF transmitter module.

When the driver presses the push button, the RF transmitter sends a radio frequency signal to nearby traffic signal controllers.

Features

- Wireless communication
- Long transmission range
- Low power consumption

The transmitter operates at frequencies such as **433 MHz**, which are commonly used in

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wireless communication modules.

4.4 RF Receiver Unit

The RF receiver unit is installed at the traffic signal controller. It continuously listens for incoming RF signals transmitted by emergency vehicles.

When the receiver detects a signal, it converts the RF signal into an electrical signal and sends it to the priority logic circuit.

Functions of RF Receiver

- Receive wireless signals
- Convert RF signals into digital signals
- Trigger traffic signal control circuit

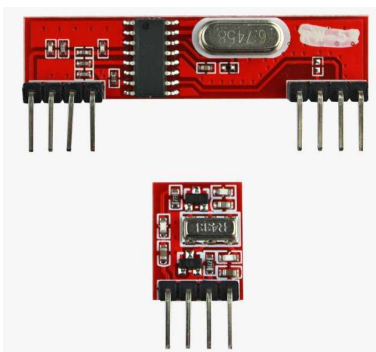


Fig 4.4:RF Transmitter & Receiver

4.5 Priority logic Circuit

The priority logic circuit determines which emergency vehicle receives priority at the intersection.

Vehicle	Priority Level
Ambulance	Highest
Fire Truck	Medium
Police Vehicle	Lower

Table 4.5: Priority Logic Circuit

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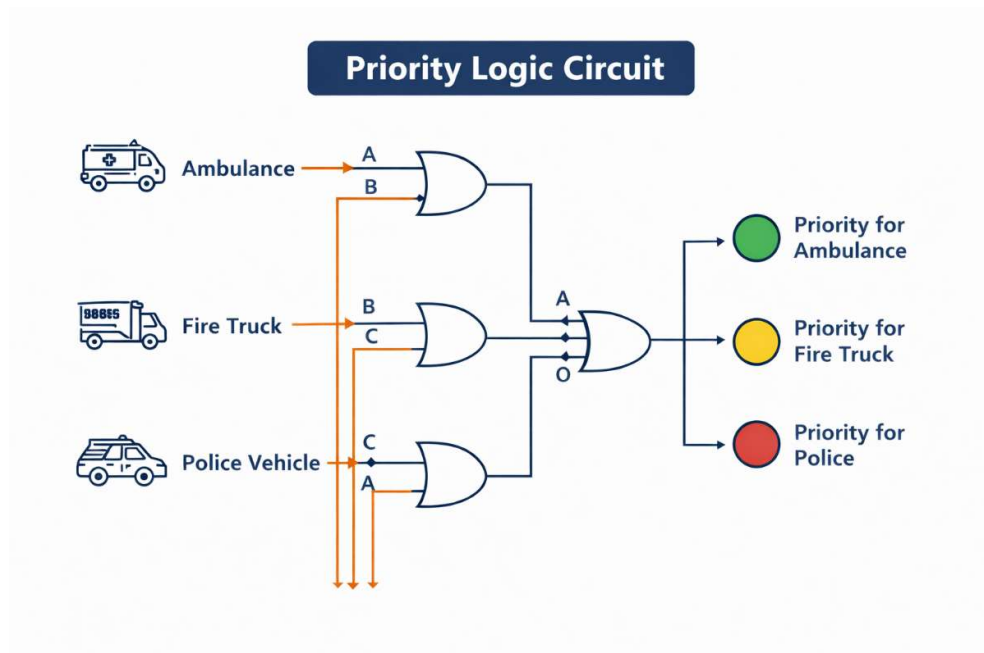


Fig 4.5 Priority Logic Circuit

4.6 Automatic Reset Circuit

The automatic reset circuit ensures that traffic signals return to normal operation after the emergency vehicle passes.

This circuit uses a **555 Timer IC** configured in monostable mode. When an emergency signal is received, the timer starts counting. After the preset time interval expires, the circuit resets the traffic signal controller.

Advantages

- Prevents traffic disruption
- Automatic restoration of normal traffic signals
- Adjustable delay time

CHAPTER-V

HARDWARE DESIGN

5.1 Introduction

The proposed system is implemented using several electronic components. These components work together to detect emergency vehicle signals and control traffic lights.

5.2 RF Transmitter Module

The RF transmitter module sends wireless signals to the receiver.

Features

- Operating frequency: 433 MHz
- Low power consumption
- Compact size

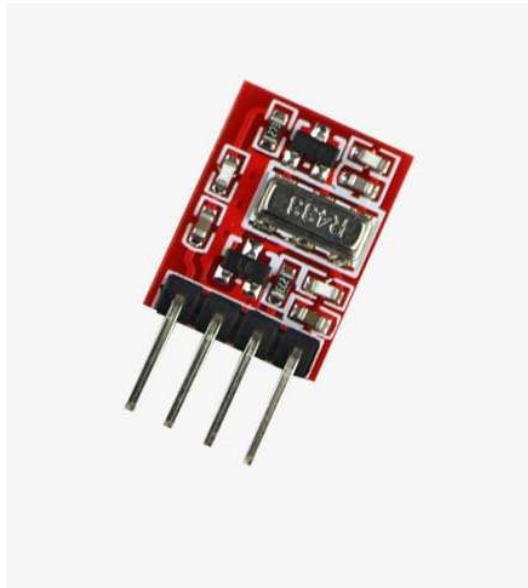


Fig 5.2 RF Transmitter Module

5.3 RF Receiver Module

The RF receiver module detects signals transmitted by the RF transmitter.

Features

- High sensitivity
- Reliable signal detection

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- Easy integration with circuits

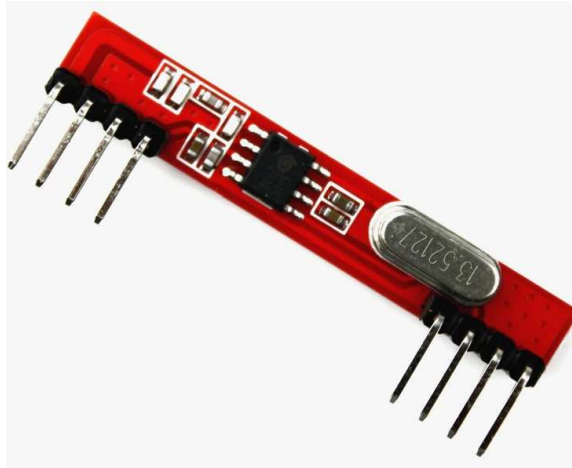


Fig 5.3 RF Receiver Module

5.4 Push Button Switch

Push buttons are used to activate the RF transmitter. Each emergency vehicle has a dedicated push button.

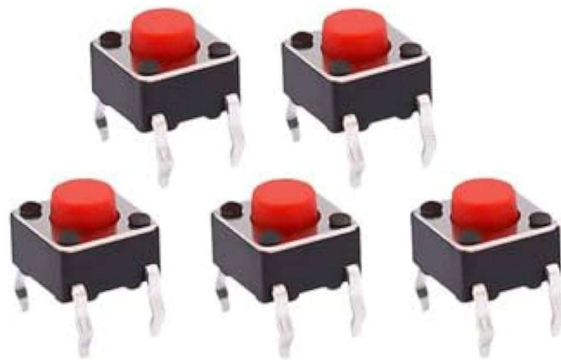


Fig 5.4: Push Button Switch

5.5 Logic Gates

Logic gates are used to implement priority control logic.

Commonly used gates include:

- AND gate
- OR gate

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- NOT gate

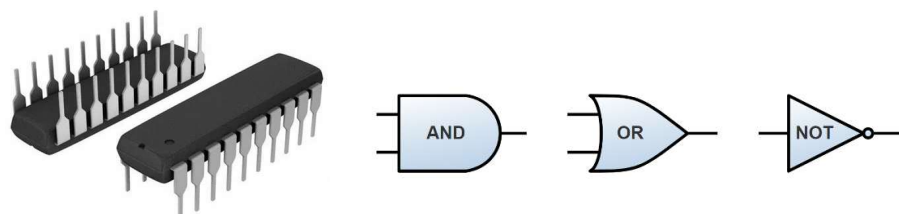


Fig 5.5: Logic Gates

5.6 555 Timer IC

The 555 Timer IC is used for generating time delays in the automatic reset circuit.

Modes of Operation

- Monostable mode
- Astable mode
- Bistable mode

In this project, the **monostable mode** is used.

555 Timer Circuit using NE555 IC

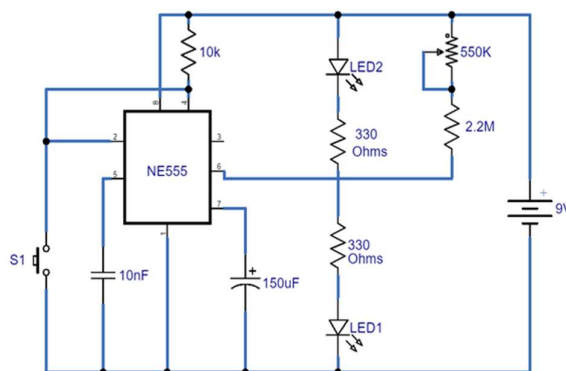


Fig 5.6: 555 Timer IC

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5.7 Traffic Light LED

Traffic light LEDs represent the traffic signals.

- Red LED – Stop
- Yellow LED – Prepare to stop
- Green LED – Go



Fig 5.7:Traffic Light Module

5.8 Power Supply Unit

The circuit requires a stable power supply to operate correctly. A regulated DC power supply is used to power the system.



Fig 5.8:Power Supply Unit

CHAPTER-VI

EXPERIMENTAL RESULTS

6.1 Introduction

After designing the circuit, experimental tests were conducted to evaluate the performance of the proposed system.

6.2 Testing Procedure

- The RF transmitter was activated using push buttons.
- The RF receiver detected the transmitted signals.
- The priority logic circuit processed the signals.
- The traffic signal LEDs were activated accordingly.

6.3 Experimental Observations

Test Condition	Result
Ambulance signal	Immediate green signal
Fire truck signal	Medium priority
Police vehicle signal	Lower priority
No emergency signal	Normal traffic operation

Table 6.3: Experimental observations

6.4 Performance Analysis

The experimental results show that the proposed system successfully detects emergency vehicle signals and provides priority traffic control.

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The RF communication system provides reliable signal transmission and detection.

CHAPTER-VII

ADVANTAGES & APPLICATIONS

7.1 Advantages of the proposed method

- Faster emergency response time
- Reduced traffic congestion
- Low cost implementation
- Reliable wireless communication
- Simple circuit design
- Easy installation and maintenance

7.2 Applications

- Smart city traffic management systems
- Emergency vehicle traffic control
- Disaster management systems
- Military convoy traffic control
- Airport vehicle management
- Industrial traffic control

CHAPTER-VIII

CONCLUSION & FUTURE SCOPE

8.1 Summary of work

This project presents the design and implementation of an RF-based emergency vehicle traffic signal control system. The system allows emergency vehicles to transmit signals to traffic signal controllers and obtain priority passage at intersections.

8.2 Conclusion drawn from results

The experimental results demonstrate that the proposed system effectively detects emergency vehicle signals and provides priority traffic control. The RF communication system ensures reliable wireless transmission.

8.3 Limitations of the current system

- Limited communication range
- Prototype implementation only
- Cannot detect vehicle location automatically

8.4 Future scope

- Integration with IoT traffic systems
- GPS-based emergency vehicle detection
- AI-based traffic management
- Integration with smart city infrastructure

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APPENDIX

Appendix A — List of Components Used

S.No	Component Name	Specification	Quantity
1	RF Transmitter Module	433 MHz	1
2	RF Receiver Module	433 MHz	1
3	Push Button Switch	Normally Open	3
4	Logic Gate ICs	AND / OR Gates	2
5	555 Timer IC	NE555	1
6	LEDs	Red, Yellow, Green	3
7	Resistors	220 Ω	3
8	Capacitors	10 μ F	1
9	Power Supply	5V DC	1
10	Breadboard / PCB	General Purpose	1

Table A:List of components

Appendix B — Circuit Diagram

The circuit diagram of the proposed system includes the following sections:

1. RF Transmitter Circuit
2. RF Receiver Circuit
3. Priority Logic Circuit
4. Traffic Signal LED Circuit
5. Automatic Reset Circuit using 555 Timer

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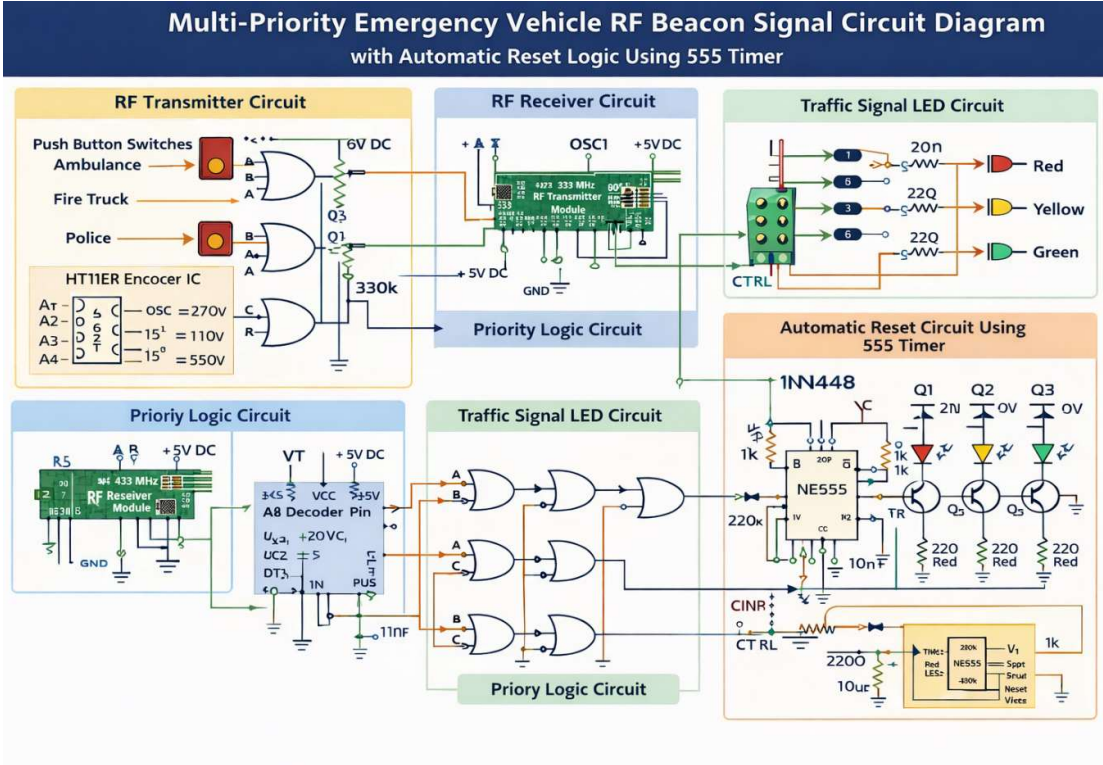


Fig B: Circuit Diagram

Appendix C — Truth Table for Priority Logic

The priority logic circuit determines which emergency vehicle gets priority at the traffic signal.

Ambulance	Fire Truck	Police	Output Priority
1	0	0	Ambulance
0	1	0	Fire Truck
0	0	1	Police
1	1	0	Ambulance
1	0	1	Ambulance
0	1	1	Fire Truck

Table C:Truth Table for Priority Logic

Appendix D – Timing Waveform

The timing waveform illustrates the operation of the traffic signals when an emergency vehicle signal is received.

During normal operation, the traffic signal follows a standard sequence:

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Red → Yellow → Green

When an emergency vehicle signal is received, the corresponding signal immediately switches to green.

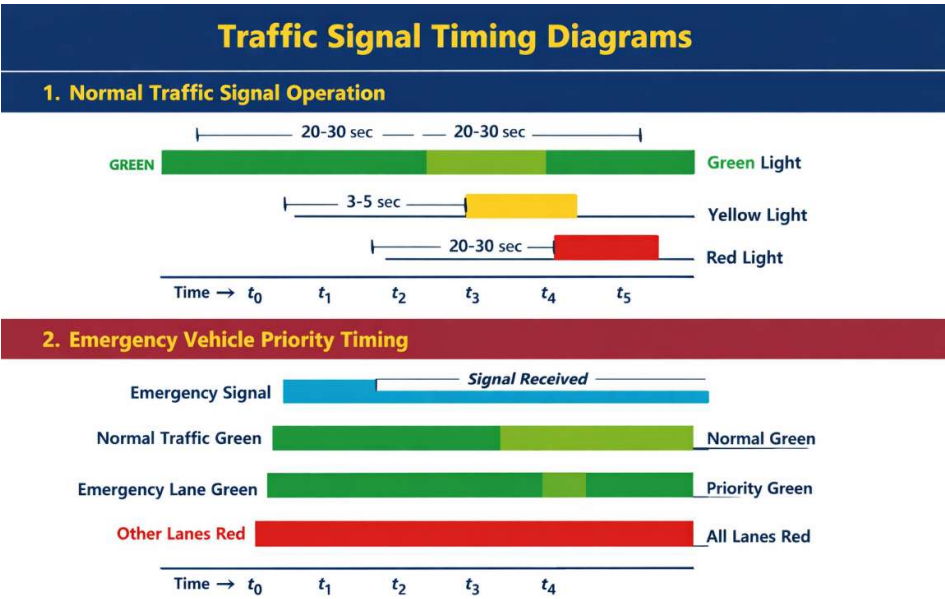


Fig D:Traffic Signal Timing Waveform

Appendix E – Hardware Setup

The hardware prototype of the system was implemented using RF modules, logic gate circuits, and traffic light LEDs. The RF transmitter was activated using push buttons representing emergency vehicle signals.

✦ **Insert Figure E.1 here:**

Hardware Prototype of Emergency Vehicle RF Beacon System